

# **Questions TopicWise JAVA**

## **OOPs**

**Q.1 What do you mean by Inheritance ? What are the advantages of Inheritance ?**

**Q.2 Which class is by default parent class of all java classes ? why ?**

**Q.3 How many types of Inheritance are possible explain each and which type of Inheritance not supported in java ?**

**Q.4 What do you mean by multiple Inheritance ?**

**Q.5 Why java doesn't support multiple inheritance?**

**Q.6 What is an ambiguity problem or diamond problem regarding inheritance in java ?**

**Q.7 Why does the ambiguity problem not arise in the case of interfaces in java ?**

**Q.8 What do you mean by a HAS-A relationship ?**

**Q.9 What do you mean by composition and aggregation in JAVA ?**

**Q.10 Give practical example of composition and aggregation in JAVA ?**

**Q. 11 What do you mean by method signature ?**

**Q.12 What do you mean by method overloading ?**

**Q. 13 What do you mean by method overriding ?**

**Q. 14 What do you mean by method hiding ?**

**Q. 15 Explain the differences between method overriding and method hiding in java ?**

**Q. 16 Explain the differences between method overloading and method overriding in java ?**

**Q. 17 What do you mean by polymorphism ? How many types of polymorphism are there ?**

**Q.20 What is the main purpose of the Constructor ?**

**Q. 23 What do you mean by default constructor ?**

**Q. 24 What are the differences between super(), this() and super,this ?**

**Q. 26 What do you mean by object type casting ?**

**Q. 27 Explain rules for object type casting ?**

**Q.28 What do you mean by Data hiding, Abstraction, Encapsulation in Java ?**

**Q.29 What is the purpose of the static block ?**

**Q.30 What is the purpose of the instance block ?**

**Q.31 What do you mean by coupling ?**

**Q.32 What do you mean by Cohesion ?**

# **Language Fundamentals**

**Q.1 What do you mean by Identifiers ?**

**Q.2 Explain Rules of Java Identifiers ?**

**Q.3 Difference between float and double data type ?**

**Q.4 Why char data types takes 2 byte of memory in java ?**

**Q.5 What are the advantages and disadvantages of array ?**

**Q.7 What are the differences between instance and static variables ?**

**Q.8 What do you mean by the var-arg method ?**

**Q.9 What are the advantages of the var-arg method ?**

**Q.10 What changes can we perform on main() ?**

**Q.12 What do you mean by command line arguments ?**

# **Exception Handling**

**Q.1\_What do you mean by exception ?**

**Q.2 What is the main objective of Exception Handling ?**

**Q.3 What is the meaning of Exception Handling ?**

**Q.5 Which is the root class for Java Exception Hierarchy and why ?**

**Q.6 What are the differences between checked exceptions & unchecked exceptions ?**

**Q.7 What are the differences between partially checked exceptions & fully checked exceptions ?**

**Q.8 What is the purpose of try and catch block ?**

**Q.9 Explain about methods to print exception information to the console ?**

**Q.10 What is the main purpose of try with multiple catch blocks ?**

**Q.11 What is the main purpose and speciality of finally block ?**

**Q.13 What is the purpose of the throw keyword ?**

**Q.14 What is the purpose of the throws keyword ?**

**Q.15 How many types of exceptions are available in java ?**

**Q.16 Explain any 5 exceptions of java ?**

# **Declaration & Access Modifiers**

**Q.1\_How many classes a java program can contain ?**

**Q.2 What do you mean by fully qualified name and what is the problem in using a fully qualified name ?**

**Q.3 What is the purpose of import ?**

**Q.4 Explain types of import statement ?**

**Q.6 What do you mean by static import ?**

**Q. 7 What are the advantages and disadvantages of using static import ?**

**Q.8 What do you mean by package ?**

**Q.9 What do you mean by public modifier ?**

**Q.10 What do you mean by private modifier ?**

**Q.11 What do you mean by <default> modifier ?**

**Q.12 What do you mean by protected modifier ?**

**Q.13 What do you mean by final modifier ?**

**Q.14 Explain abstract class and abstract method ?**

**Q.15 What is the purpose of strictfp modifier ?**

**Q.16 What is the purpose of native modifier ?**

**Q.17 How many modifiers are applicable on outer class ?**

**Q.18 Which modifiers are applicable for local variables ?**

**Q.19 Which modifiers are applicable only for method ?**

**Q.20 Which modifiers are applicable only for variables ?**

**Q.21 On Constructors which modifiers are applicable?**

# **Interface**

**Q.1\_What do you mean by interface ?**

**Q.2 Explain about extends and implements keywords ?**

**Q.3 What is the purpose of import ?**

**Q.4 Explain about interface methods and interface variables ?**

**Q.5 Multiple Inheritance possible in java or not ?**

**Q.6 Why is multiple Inheritance not supported in classes but supported in Interfaces ?**

**Q.7 How many Interfaces a Java class can implement simultaneously ?**

**Q.8 What do you mean by marker Interface ?**

**Q.9 What do you mean by Adapter class ?**

**Q.10 What are the differences between abstract class & Interface ?**

**Q.11 What do you mean by default method in Interface ?**

**Q.12 What do you mean by static method in Interface ?**



# **Java.lang package**

**Q.1 Explain about Object class?**

**Q.2 Explain about toString() ?**

**Q.3 What do you mean by hashCode ?**

**Q.4 Explain about equals() ?**

**Q.5 Explain the relation between == operator and equals() ?**

**Q.6 What are the differences between == operator and equals() ?**

**Q.7 What is the purpose of cloning ?**

**Q.8 What are the differences between shallow cloning and deep cloning ?**

**Q.9 What are the differences between String and StringBuffer ?**

**Q.10 Why is the String class object immutable ?**

**Q.11 What are the advantages and disadvantages of scp ?**

**Q.12 What are the differences between equals() and equalsIgnoreCase() ?**

**Q.13 What are the differences between StringBuffer and StringBuilder ?**

**Q.14 What do you mean by wrapper classes ?**

**Q.15 Which wrapper classes are not direct child classes of the Object class ?**

**Q.16 Which wrapper classes are direct child classes of the Object class ?**

**Q.17 What do you mean by auto boxing & auto unboxing ?**

# **Garbage Collection**

**Q.1\_What do you mean by Garbage Collection ?**

**Q.2 When an object is said to be eligible for GC ?**

**Q.3 Explain ways to make an object eligible for GC ?**

**Q.4 How can we request JVM to run GC ?**

**Q.5 Explain the purpose of finalize() ?**

**Q.6 What do you mean by memory leaks ?**

# **Collection Framework**

**Q.1\_What are the advantages and disadvantages of array ?**

**Q.2 What are the advantages of collection ?**

**Q.3 What are the differences between array & collection ?**

**Q.4 What are the differences between add() and addAll()  
method of Collection interface ?**

**Q.5 What is the purpose of retainAll() of Collection interface ?**

**Q.6 What are the differences between contains() and  
containsAll() method of Collection interface ?**

**Q.7 What are the differences between Collection & Collections  
?**

**Q.8 Explain the properties of ArrayList ?**

**Q.9 What is the relation between current capacity & new  
capacity of ArrayList ?**

**Q.10 What are the differences between ArrayList & Vector ?**

**Q.11 When ArrayList is the best choice ?**

**Q.12 Explain the properties of LinkedList ?**

**Q.13 When LinkedList is the best choice ?**

**Q.14 Explain the properties of Vector ?**

**Q.15 Explain the properties of Stack ?**

**Q.16 What are the differences between pop() and peek() of Stack class ?**

**Q.17 What do you mean by cursors & how many cursors are available in Java ?**

**Q.18 Explain the properties of the Enumeration cursor ?**

**Q.19 What are the limitations of the Enumeration cursor ?**

**Q.20 Explain the properties of the Iterator cursor ?**

**Q.21 What are the limitations of the Iterator cursor ?**

**Q.22 Explain the properties of the ListIterator cursor ?**

**Q.23 What are the limitations of the ListIterator cursor ?**

**Q.24 What are the differences among all the 3 cursors ?**

**Q.25 Explain the properties of HashSet ?**

**Q.27 What are the differences between HashSet & LinkedHashSet ?**

**Q.28 Explain the properties of TreeSet ?**

**Q.29 What are the differences between Comparable & Comparator ?**

**Q.30 What is the purpose of the Map interface ?**

**Q.32 Explain the properties of HashMap ?**

**Q.33 What are the differences between HashMap & Hashtable ?**

**Q.34 What are the differences between HashMap & LinkedHashMap ?**

**Q.35 What are the differences between HashMap & IdentityHashMap ?**

**Q.36 What are the differences between HashMap & WeakHashMap ?**

**Q.37 Explain the properties of TreeMap ?**

**Q.38 What is the main purpose of Generics ?**

**Q.39 What are the differences between ArrayList l=new ArrayList() and ArrayList<String> l=new ArrayList<String>() ?**

# **Multi-Threading**

**Q.1 What do you mean by Multi-Tasking ?**

**Q.2 What are the differences between process based multitasking and thread based multitasking ?**

**Q.3 By How many ways can we create a Thread in Java ?**

**Q.4 What do you mean by Thread Scheduler ?**

**Q.5 What are the differences between t.start() & t.run() ?**

**Q.6 What is the importance of Thread class start() ?**

**Q.9 Explain Thread life cycle in Java ?**

**Q.10 How can we set and get the name of a thread in Java ?**

**Q.11 How can we get current executing thread object in Java ?**

**Q.12 What do you mean by Thread Priority In Java ?**

**Q.13 Explain the concept of default priority ?**

**Q.14 Explain the purpose of yield() ?**

**Q.15 Explain the purpose of join() ?**

**Q.16 Explain the purpose of sleep() ?**

**Q.17 What do you mean by synchronization ?**

**Q.18 What are the advantages & disadvantages of synchronization ?**

**Q.19 What do you mean by object level lock ?**

**Q.21 What is the purpose of synchronized block ?**

**Q.22 What do you mean by daemon thread ?**

**Q.23 What is the default daemon nature ?**

**Q.24 What do you mean by Inter Thread Communication ?**

**Q.25 What is the difference between notify() and notifyAll() ?**

**Q.26 What do you mean by deadlock ?**

**Q.27 What are the differences between deadlock & starvation ?**